
Why Language Models Transfer to Time Series

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 TODO

2 **1 Introduction**

3 **2 Related work**

4 **Time-series foundation models (TSFMs).** Models designed as general-purpose time series fore-
5 casters, trained on Large-scale multi-domain data

6 1. Chronos []

7 2. Lag-Llama []

8 3. MOMENT []

9 4. TimeGPT []

10 **LLM-to-time-series transfer.** Work that repurposes pretrained language models for time series,
11 primary related work

12 1. "Are Language Models Actually Useful for Time Series Forecasting?" [Tan et al., 2024]

13 2. Time-LLM: Time Series Forecasting by Reprogramming Large Language Models [Jin et al.,
14 2023]

15 3. Using Pre-trained LLMs for Multivariate Time Series Forecasting [Wolff et al., 2025]

16 4. isionTS: Visual Masked Autoencoders Are Free-Lunch Zero-Shot Time Series Forecasters
17 [Chen et al., 2024]

18 **Time Series Tokenization and Representation; LLM2TS continued**

19 1. A Time Series is Worth 64 Words [Nie et al., 2022]

20 2. Small Vocabularies, Big Gains [Roger et al., 2025]

21 **The Platonic Representation Hypothesis? Cross-Modal Geometry?** Theoretical/empirical work
22 on representation convergence to explain why might representations transfer across modalities ->
23 Language to TS

24 1. The Platonic Representation Hypothesis [Huh et al., 2024]

25 2. Harnessing the Universal Geometry of Embeddings [Jha et al., 2025]

26 3. Training Large Language Models to Reason in a Continuous Latent Space [Hao et al., 2024]

27 **Scaling Laws and Efficient Adaptation** How (do we) do models scale and fine-tuning strategies
 28 interact with transfer

- 29 1. Scaling Laws for Transfer [Hernandez et al., 2021]
- 30 2. Studies on when scaling hurts? []
- 31 3. Parameter-efficient fine-tuning comparisons, LoRA vs Full vs Adapters []

32 The findings that pretrained 0.6B matches random 4B implies a transfer multiplier of $\sim 8\times$ in effective
 33 parameters; `io_only` regime (embedding + `lm_head` only) indicate if the transfer is fully captured by
 34 the i/o layers $\rightarrow$ PEFT/LoRA

35 3 Establishing Transfer Learning is Real

36 **Setup.** All models share the Qwen3 architecture. We train pairs of **FT** (pretrained $\rightarrow$ finetuned on
 37 TS) and **RI** (random init $\rightarrow$ trained on TS) across multiple scales (0.6B, 1.7B, 4B) using the same
 38 TS training procedure (uniform binning, 512 bins, context length 512). All models are evaluated on
 39 GiftEval (46 datasets).

40 **Main result.** FT consistently outperforms RI across scales and datasets.

Table 1: ND[0.5] on GiftEval (lower is better). FT outperforms RI at all scales.

Model	0.6B	1.7B	4B
FT (pretrained + TS)	TODO	TODO	TODO
RI (random + TS)	TODO	TODO	TODO
Δ (FT – RI)	TODO	TODO	TODO

41 **Scaling interaction.** Pretrained 0.6B matches or exceeds random-init 4B, implying a transfer
 42 multiplier of $\sim 8\times$ in effective parameters.

43 **Finetuning regime comparison.** We compare full finetuning, LoRA, and IO-only (embedding +
 44 LM head only).

45 4 The Shared Manifold Hypothesis

46 Having established the existence of transfer learning from LLMs to time series in Sec. 3, we now
 47 present a geometric explanation for why LLMs are such adept time series forecasters. Due to the
 48 recursive structure and statistical patterns exhibited by language [?], we theorize that pretrained
 49 language model representations already contain native temporal structures well suited for adaptation
 50 to time series data. In this section, we present several experiments demonstrating that LLMs do
 51 possess a manifold that is already compatible with time series forecasting, and that fine-tuning does
 52 not need to create new directions, it simply identifies and collapses time series representations onto
 53 the best pre-existing directions.

54 4.1 A Linear Probe Reveals Latent Temporal Structure

55 In these experiments, we learn an linear mapping, $\hat{y}_i \in \mathbb{R}^T$, provided in Eq. 1, from a frozen,
 56 pre-trained language model’s hidden states. For each WikiText sequence, hidden states are extracted
 57 from all 28 layers and concatenated per timestep, $\mathbf{h}_{i,t} = [\mathbf{h}_{i,t}^{(0)}; \dots; \mathbf{h}_{i,t}^{(27)}] \in \mathbb{R}^D$, where D is 28,672.
 58 The per-timestep linear map then produces a predicted sequence that is z-normalized to zero mean
 59 and unit variance. Using an EM-style nearest-neighbor algorithm, each prediction is matched with its
 60 closest time series from GiftEval. Fig. 1b(a) shows plots of three of these predictions (blue) along
 61 with their closest GiftEval matches (green), on a set of data not included in the training of the linear
 62 layer. The mapping is optimized to minimize MSE to these assignments, with a spectral diversity
 63 penalty, λ , to prevent collapse.

$$\hat{y}_{i,t} = \mathbf{w}^\top \mathbf{h}_{i,t} + b \quad (1)$$

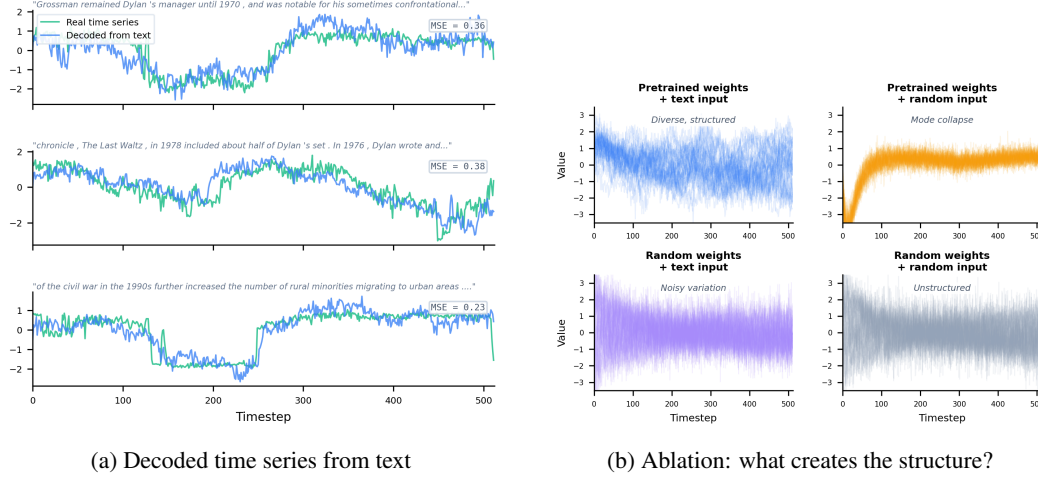


Figure 1: **Pretrained hidden states contain time-series-compatible structure.** A single linear map $\hat{y}_t = \mathbf{w}^\top \mathbf{h}_t + b$ is trained on frozen LLM hidden states via EM-style nearest-neighbor matching to a bank of 10,000 real time series (no paired data). **(a)** Three decoded outputs (blue) overlaid with their nearest real time series (green). Input text shown above each plot. $\text{MSE} = 0.23\text{--}0.38$; full distribution in Appendix. **(b)** Ablation crossing learned weights vs. random init with text vs. random input. *Top-left*: pretrained + text—diverse, structured outputs (301 unique matches). *Top-right*: pretrained + random tokens—mode collapse to one shape (manifold exists but is not traversed). *Bottom-left*: random weights + text—noisy variation without structure. *Bottom-right*: random weights + random tokens—unstructured. Only pretrained weights *and* meaningful text together produce diverse temporal signals.

64 Out of 1920 wiki text samples, the linear map produces 686 unique matches to one of the 10,000
 65 real time series. The map generalizes to 439 unique held-out WikiText text sequences out of 1000,
 66 strongly suggesting that the pre-trained language model’s representation space already contains
 67 directions whose projections resemble temporal signals.

68 The fact that linear probing can extract realistic time-series from language model hidden states has
 69 direct implications for fine-tuning these language models to predict time-series. Let $\mathcal{M} \subset \mathbb{R}^D$ denote
 70 the manifold of hidden-state trajectories produced by the pre-trained language model processing
 71 text. The experiments demonstrate the existence of a direction $\mathbf{w} \in \mathbb{R}^D$ such that the distribution of
 72 projected trajectories, $\{\mathbf{w}^\top \mathbf{h}_t\}_{t=1}^T$ has significant overlap with the distribution of real time series,
 73 P_{TS} . If we define $\mathbf{h} * t$ as the hidden state of the pretrained model due to its autoregressive training,
 74 the model implements a deterministic update shown in Eq. 2 and learns to predict x_{t+1} from $\mathbf{h}_t$. In
 75 practice, this means that the model has learned structured dynamics over the hidden states and given
 76 $\mathbf{h} * t$, it can reliably produce the next state $\mathbf{h} * t + 1$ along trajectories induced by natural text.

$$\mathbf{h}_{t+1} = F(\mathbf{h} * t, x * t + 1) \quad (2)$$

77 The linear probing experiments demonstrate that a projection direction, $\mathbf{w}$, resembling a real-world
 78 time series can indeed be found within the frozen language models representation space. Because the
 79 LLM is trained to predict the next token x_{t+1} by evolving its internal state, the function F in Eq. 2
 80 acts as a latent dynamical system for the temporal manifold. The model must implicitly predict $\mathbf{h}_{t+1}$
 81 to make autoregressive predictions so it already possess the ability to propagate these signals through
 82 time. Once obtained, $\mathbf{h} * t + 1$ can be combined with the pre-existing direction $\mathbf{w}$ and applied to
 83 Eq. 1 to produce $y_{t+1} = \mathbf{w}^\top \mathbf{h}_{t+1}$. This confirms that the pre-trained language model can propagate
 84 hidden states along trajectories consistent with real-world time series.

85 In summary, fine-tuning language models for time series does not require learning the dynamics
 86 from scratch, it is a low-dimensional alignment problem that requires selecting and scaling existing
 87 directions in representation space.

4.2 this needs a new section or appendix relocation

The latent temporal structure exhibited above could have a variety of explanations which we attempt to disentangle by crossing model weights (pretrained vs. randomly initialized, same architecture) with input (WikiText vs. random tokens). A clear interaction emerges, *pre-trained weights* create a structured, low-dimensional manifold, but random tokens fail to traverse it, Tab. 2 shows that *rand+PT* collapses to just four unique outputs. *Meaningful text* provides variation, but without pretrained weights, this variation is unstructured noise (Tab. 2 *text+RandInit*). Only the combination produces diverse, high-quality temporal signals. In this case, the pretrained manifold provides the geometry, and text input drives its diverse traversal.

Table 2: 2×2 ablation. *Unique*: distinct real TS matched. *NN*: mean nearest-neighbor MSE. Low NN with low Unique indicates mode collapse, not quality. We need to fix the NN because its not normalized by the number of matches

Condition	Train Unique	Held-Out Unique	Held-Out NN	Clusters
text + PT	686/1920	439/1000	0.717	14/20
text + RandInit	54/1920	73/1000	0.541	13/20
rand + PT	4/1920	1/1000	0.293	4/20
rand + RandInit	99/1920	138/1000	0.741	14/20

4.3 The Projected Dynamics Forecast Beyond the Context

The linear decoding result shows shape-level similarity between projected hidden states and real time series. We test whether this extends to *predictive* structure: given the first half of a time series, can a retrieved text projection forecast the second half?

Method. We take 500 real time series from GiftEval and observe only the first 256 of 512 timesteps. Among 10,000 WikiText projections, we retrieve the one with lowest MSE on the observed first half and use that projections second half as the forecast. The baseline is last-value carry-forward *i.e.* repeating the final observed value.

Result. Across all 500 queries, the retrieval-based forecast achieves $MSE = 1.91$ versus last-values 2.27 (16% lower). The retrieval wins in 37% of cases; when it wins, the advantage is large (median $4\times$ lower MSE), because the retrieved projection captures trends and level shifts that last-value misses entirely. Figure 2 shows the best-case example from our evaluation set.

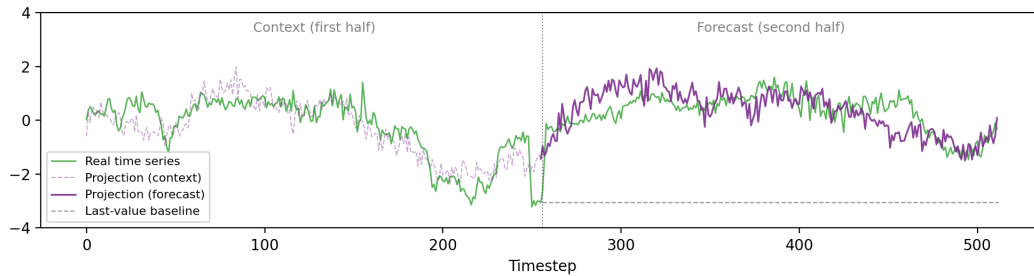


Figure 2: Forecasting example with largest gap compared to baseline from 500 evaluation queries. The first 256 timesteps are used to retrieve the closest WikiText projection; the projection’s second half (purple) forecasts the continuation. The retrieved projection captures the level shift and subsequent trend ($MSE = 0.42$), while last-value carry-forward (gray dashed) is stranded at the final context value ($MSE = 11.7$).

4.4 Pretraining Creates Coherent Gradient Signal for the TS Objective

details in the md file <https://hackmd.io/VIS4nmT6Q2ms66mMUXTgWQ>

figures/sec4/fig_headline_7.png

Figure 3: .

111 The manifold structure has direct consequences for optimization. We compute per-example gradients
 112 of the TS loss at the pretrained versus random initialization, measuring whether individual training
 113 examples provide coherent or contradictory optimization signals.

114 **Method.** For 30 individual TS sequences, we compute the gradient $\mathbf{g}_i = \nabla_{\theta} L_i$ restricted to layers
 115 6–10 (78.7M parameters) at both initializations. We measure pairwise cosine similarity $\cos(\mathbf{g}_i, \mathbf{g}_j)$
 116 across all example pairs, and the signal-to-noise ratio $\text{SNR} = \|\bar{\mathbf{g}}\|/\text{mean}(\|\mathbf{g}_i - \bar{\mathbf{g}}\|)$.

Table 3: Per-example gradient structure at the TS loss surface. At PT initialization, different TS examples produce gradients that reinforce each other. At random initialization, gradients are nearly orthogonal and cancel upon averaging.

Metric	PT init	Random init
Per-example gradient cosine	0.58	0.04
Gradient signal-to-noise ratio	1.16	0.28
Gradient norm (RMS)	0.0040	0.0004

Interpretation. At PT initialization, per-example TS gradients have cosine similarity 0.58—different time series sequences agree on which direction to adjust the weights. Batch averaging reinforces this shared signal. At random initialization, gradients are nearly orthogonal (cosine 0.04, consistent with random vectors in $\sim 78\text{M}$ dimensions), so averaging causes massive cancellation: the SNR drops to 0.28, meaning most of the per-example gradient signal is lost.

This coherence arises because the pretrained representations create consistent feature activations across different TS inputs. When the loss asks “how should this weight change?”, the answer is similar regardless of which TS sequence is used, because the pretrained features impose a shared structure on the loss landscape. At random initialization, each example activates arbitrary features, giving contradictory signals.

The effect is layer-dependent: gradient norms at PT initialization are $3\text{--}19\times$ larger than at random initialization, with the ratio increasing for deeper layers (where random weights scramble the signal most). This layer-wise gradient structure explains why mid-to-deep layers benefit most from pretrained initialization (Section 5.2).

Curvature anisotropy. To characterize the *shape* of the loss landscape, we compute the curvature $\mathbf{v}^\top H \mathbf{v}$ along 200 random unit vectors $\mathbf{v}$ in layers 6–10 parameter space using exact Hessian-vector products (Pearlmutter trick), and compare this to the curvature along the gradient direction $\hat{\mathbf{g}}$. At PT initialization, the gradient direction has curvature 232, while the mean curvature across random directions is 9.6×10^{-5} —a ratio of 2.4×10^6 . At random initialization, the gradient direction has curvature -0.08 (near-zero, slightly negative), barely distinguishable from random directions (mean 8.4×10^{-7} , ratio $\sim 80\times$). The PT landscape is thus *massively anisotropic*: SGD’s natural descent direction is aligned with one of the very few high-curvature directions in a $\sim 78\text{M}$ -dimensional space, while the vast majority of directions are flat. At random initialization, the landscape is nearly isotropic—all directions, including the gradient, have negligible curvature.

5 Finetuning Projects the Pretrained Manifold onto Time Series

5.1 The Weight Changes are Low-Rank

Table 4: Finetuning regime comparison on GiftEval ND[0.5]. TODO: fill from Alexis’s experiments.

Method	Params changed	ND[0.5]	% of full FT
Full finetuning	100%	TODO	100%
LoRA ($r = 16$)	TODO%	TODO	TODO%
LoRA ($r = 4$)	TODO%	TODO	TODO%
IO-only (emb + head)	TODO%	TODO	TODO%
Random init (full)	100%	TODO	—

Interpretation. Only a low-rank perturbation of the pretrained weights is needed to adapt the model from language to time series. This means the bulk of the pretrained computation is preserved—finetuning adjusts a small subspace of directions.

5.2 Finetuning Selects Useful Directions

Catastrophic forgetting as geometric compression. Text $\rightarrow$ FT collapses to effective rank 2.5 (from 150 in PT). Finetuning traded ~ 140 text-useful dimensions for ~ 12 TS-useful dimensions.

Method: effective rank across training. To track how representational geometry evolves during training, we compute the effective rank of hidden-state covariance matrices at every saved checkpoint. For each checkpoint and each of the 28 transformer layers, we pass 10,000 time-series windows (length 512, drawn from the GiftEval validation split) through the model, collecting hidden states at every position except position 0 (excluded to avoid the attention-sink artifact). For FT and FT_{IO} we additionally pass 10,000 WikiText-103 sequences to track text representation quality.

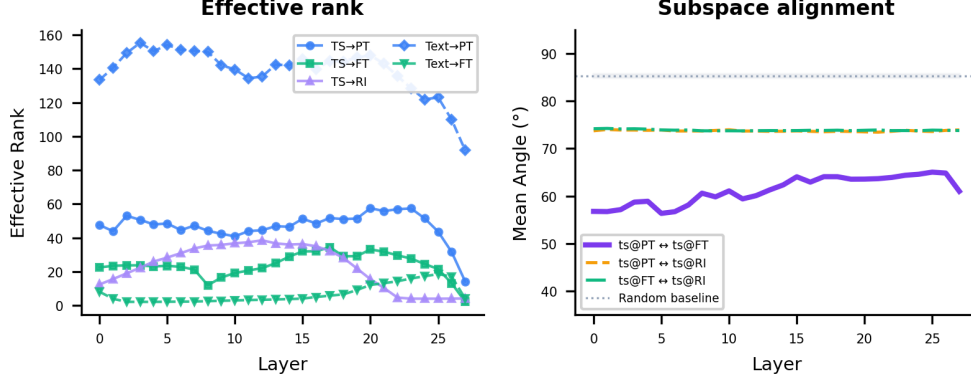


Figure 4: **Geometric evidence for direction reuse.** (Left) Effective rank across layers. FT compresses TS representations to ~ 12 dimensions (green solid), far below PT’s ~ 44 (blue solid) and RI’s ~ 35 (purple solid). Text→FT collapses to ~ 2.5 (catastrophic forgetting). (Right) Subspace alignment: mean principal angle between top-10 PCs. $\text{ts@PT} \leftrightarrow \text{ts@FT}$ (purple) stays well below the random baseline (85.3°), confirming FT reuses PT’s directions. Tables 6 and 7 in Appendix.

At each layer we accumulate the centered covariance matrix $\Sigma = \frac{1}{N} \sum_i \mathbf{h}_i \mathbf{h}_i^\top - \bar{\mathbf{h}} \bar{\mathbf{h}}^\top$ in float64 over all N hidden-state vectors ($N \approx 130,000$ per batch), then compute its eigendecomposition. The effective rank is $\text{erank}(\Sigma) = \exp(-\sum_i p_i \log p_i)$, where $p_i = \lambda_i / \sum_j \lambda_j$ are the normalized eigenvalues [Roy and Vetterli, 2007]. This equals 1 when all variance concentrates on a single direction and equals the ambient dimension d when variance is uniformly spread.

We evaluate all checkpoints for three training regimes: FT (full finetuning from PT, 15 checkpoints from step 1 to 10,000), FT_{IO} (input/output layers only, 17 checkpoints from step 1 to 33,000), and RI (randomly initialized, 14 checkpoints from step 1 to 8,192). The pretrained model (PT) serves as a fixed baseline. Checkpoints are processed 4 at a time across 4 GPUs.

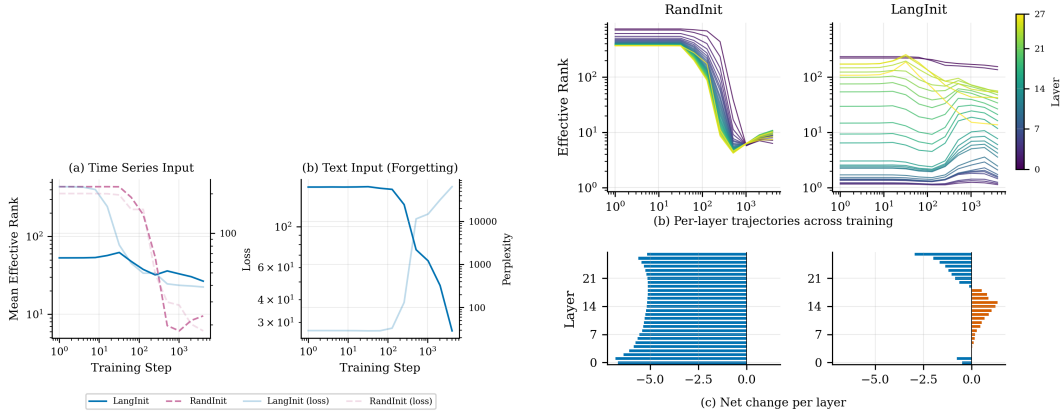


Figure 5: **Effective rank dynamics during training.** (a) Mean effective rank across all 28 layers. Left: on time-series input, RandInit starts near-isotropic (~ 440) and collapses to ~ 10 within 500 steps; LangInit declines more gradually from the pretrained baseline (~ 50) to ~ 27 . Transparent lines show CRPS loss on a secondary axis. Right: on text input, LangInit’s effective rank drops from ~ 165 to ~ 25 (85% reduction) while perplexity rises, confirming catastrophic forgetting. (b) Per-layer trajectories (each line = one of 28 layers, colored by depth). RandInit layers collapse as a tight bundle; LangInit middle layers (~ 8 – 17) increase their effective rank while early and late layers decrease. (c) $\log_2$ change (final/initial) per layer. Orange = gained rank; blue = lost. RandInit is uniformly negative (-5 to -7 , i.e., 32 – $128\times$ reduction). LangInit layers 10 – 17 show positive change (up to $+1 \log_2 = 2\times$ increase)—finetuning redistributes capacity from the network’s periphery to its core.

5.3 Reuse vs. Reinvention: LangInit and RandInit Find Different Solutions

Method. To probe how each model geometrically encodes temporal structure, we pass synthetic waveforms through the network and visualize hidden-state trajectories via 2D PCA. We generate five inputs of length 512: a sine wave (period 64), a square wave (period 64), a sawtooth wave (period 64), a two-frequency signal ($\sin(2\pi t/64) + 0.5 \sin(2\pi t/17)$), and a linear trend with oscillation ($t/T + 0.3 \sin(2\pi t/80)$). Each input is independently z -score normalized, clipped to $[-5, 5]$, and uniformly binned into 1024 tokens. The tokenized sequence is passed through the model, and we extract hidden states at the target layer. PCA is fitted independently per model–input pair, since the models occupy different regions of representation space. The first 5 positions are excluded to avoid the attention-sink artifact. Trajectories are colored by input phase (position mod period / period) to reveal periodic structure. Full per-layer breakdowns for all 28 layers are provided in Figures 9 and 10 in the Appendix.

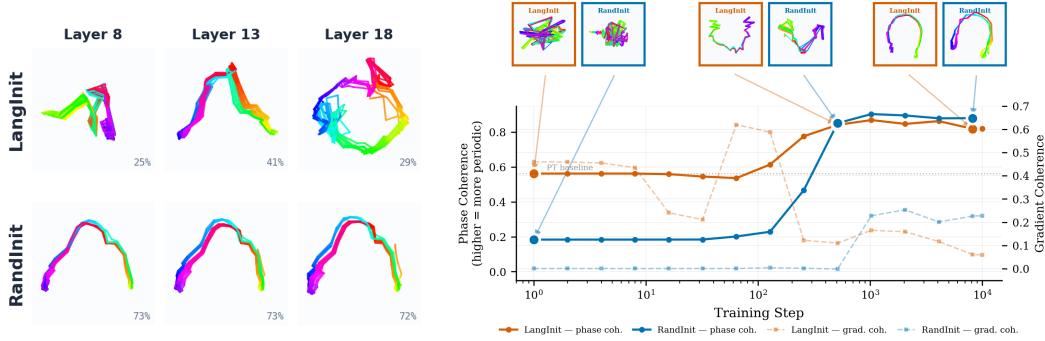


Figure 6: **Left: Sine wave representations across layers.** A sine wave (period 64) passed through LangInit (top) and RandInit (bottom) at layers 8, 13, and 18, shown as 2D PCA colored by input phase. RandInit produces clean arcs at every layer (72–73% PCA variance); LangInit produces complex, layer-varying loops (25–41%). **Right: Phase and gradient coherence across training.** Phase coherence (solid; 1 – same-phase/all-pair distance ratio, higher = more periodic) and gradient coherence (dashed) at Layer 13. Insets show t-SNE of hidden states at steps 1, 512, and 8192. LangInit inherits periodic structure from pretraining and starts high; RandInit starts near zero and undergoes an abrupt phase transition around step 256–512 where gradient coherence simultaneously spikes, indicating that the model discovers periodic geometry at the same moment it begins aligning gradients across the time series.

Figure 6 (left) shows this contrast for a single sine wave across three representative layers. RandInit converges to a uniform, clean arc at every layer, with 72–73% of variance explained by just two principal components. LangInit produces complex, layer-specific loops with only 25–41% PCA variance—the periodic structure is distributed across many more dimensions, reflecting the richer geometry inherited from pretraining. Figure 6 (right) tracks how this geometry emerges during training: LangInit inherits high phase coherence from the start, while RandInit undergoes a sharp phase transition around step 256–512 that coincides with a spike in gradient coherence.

Figure 7 extends this analysis to all five synthetic inputs at Layer 13. RandInit consistently produces clean, geometrically simple trajectories with high PCA variance, while LangInit creates more complex structures that vary by input type. Notably, both models arrive at representations that capture the same underlying temporal patterns—the t-SNE projections in Figure 8 (Appendix) confirm that the global geometries arrive at the same solution. The key difference is *interpretability*: RandInit’s representations are more readily decomposed by linear methods (PCA), whereas LangInit distributes information nonlinearly across more dimensions. The training dynamics of these representations are visualized in Figures 11 and 12 in the Appendix and confirm that pretrained models reuse the complex, high effective rank solutions confirmed in Figure 5, while RandInit first simplifies the geometry before increasing complexity (and associatively effective rank) and converging to a solution.

5.4 Interpreting the Transferred Features

Method. We train one sparse crosscoder per layer with a shared encoder and separate PT/FT decoders. The encoder maps pretrained-model activations on decimal-string time series and finetuned-

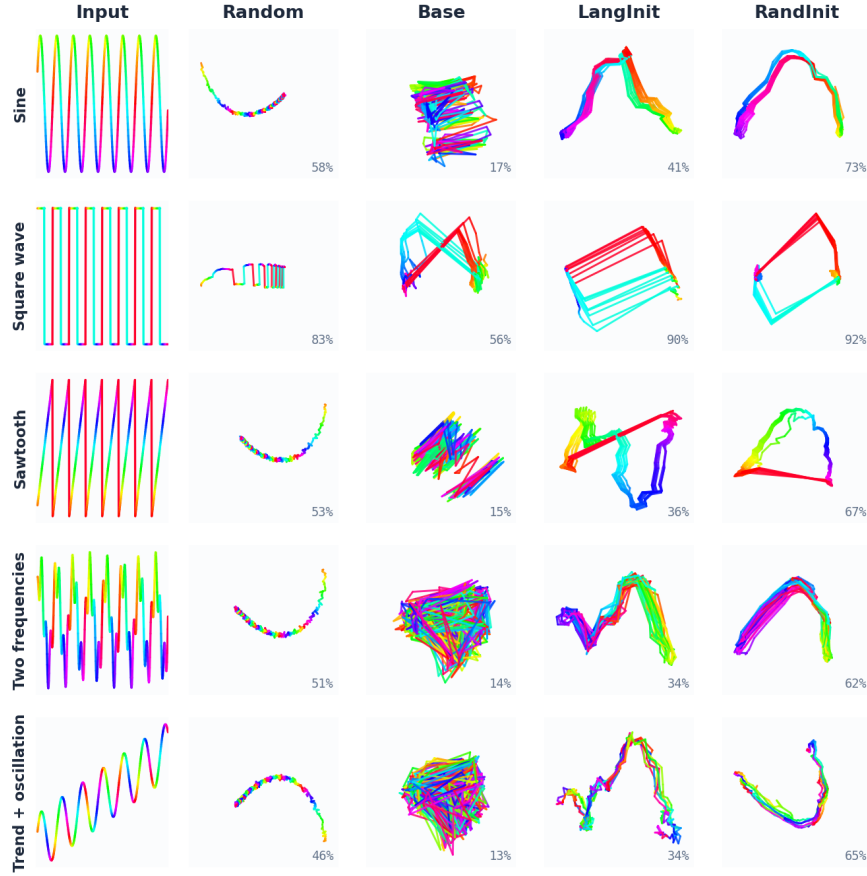


Figure 7: **Hidden-state trajectories for synthetic inputs at Layer 13 (PCA).** Each row is a different input waveform; columns show a randomly initialized untrained model (Random), the pretrained LLM (Base), the language-initialized model finetuned on time series (LangInit), and the randomly-initialized model trained on time series (RandInit). Trajectories are 2D PCA projections of hidden states, colored by input phase. Percentages show variance captured by 2 PCs. Random and Base produce unstructured trajectories. RandInit discovers clean, low-dimensional representations (62–92% PCA variance) while LangInit creates geometrically complex but structured trajectories (34–98%) that vary by input type.

196 model activations on binned time series into a shared 4,096-dimensional Top- K latent space; PT-FT
 197 features are ranked by cross-domain firing balance and probed on WikiText-103.

Table 5: Examples of shared PT-FT crosscoder features. Shared features concentrate in layers 7–10; the examples below link concrete time-series motifs to coherent WikiText genres. Full examples are in Appendix D.

Layer	Feature	Time-series motif	WikiText motif
10	1712	magnitude jumps / plateaus	measurements and unit conversions
9	2469	volatile regime changes	tropical cyclone narratives
7	3888	isolated spikes	timestamped naval battles
8	2567	missing / NaN windows	<unk> and incomplete references

198 Thus, the aligned LangInit subspace is populated by reusable primitives, not just arbitrary low-rank
 199 directions. In Appendix C, we present a preliminary causal circuit analysis that provides additional
 200 evidence: zero-ablation identifies a superadditive circuit in Layer 1 (head L1H4 $\leftrightarrow$ MLP_{L1}) that is

critical for periodic time-series prediction in FT_{IO} and, in the pretrained model, is most important for text with sequential repetitive structure.

6 Discussion and Limitations

The full picture.

Implications.

Limitations.

7 Conclusion

References

- Mouxian Chen, Lefei Shen, Zhuo Li, Xiaoyun Joy Wang, Jianling Sun, and Chenghao Liu. Visions: Visual masked autoencoders are free-lunch zero-shot time series forecasters, 2024. URL <https://arxiv.org/abs/2408.17253>.
- Arthur Conmy, Augustine N. Mavor-Parker, Aengus Lynch, Stefan Heimersheim, and Adrià Garriga-Alonso. Towards automated circuit discovery for mechanistic interpretability, 2023. URL <https://arxiv.org/abs/2304.14997>.
- Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. A mathematical framework for transformer circuits, 2021. URL <https://transformer-circuits.pub/2021/framework/index.html>.
- Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, Zhiting Hu, Jason Weston, and Yuandong Tian. Training large language models to reason in a continuous latent space, 2024. URL <https://arxiv.org/abs/2412.06769>.
- Danny Hernandez, Jared Kaplan, Tom Henighan, and Sam McCandlish. Scaling laws for transfer, 2021. URL <https://arxiv.org/abs/2102.01293>.
- Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. The platonic representation hypothesis, 2024. URL <https://arxiv.org/abs/2405.07987>.
- Rishi Jha, Collin Zhang, Vitaly Shmatikov, and John X. Morris. Harnessing the universal geometry of embeddings, 2025. URL <https://arxiv.org/abs/2505.12540>.
- Ming Jin, Shiyu Wang, Lintao Ma, Zhixuan Chu, James Y. Zhang, Xiaoming Shi, Pin-Yu Chen, Yuxuan Liang, Yuan-Fang Li, Shirui Pan, and Qingsong Wen. Time-llm: Time series forecasting by reprogramming large language models, 2023. URL <https://arxiv.org/abs/2310.01728>.
- Yuqi Nie, Nam H. Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam. A time series is worth 64 words: Long-term forecasting with transformers, 2022. URL <https://arxiv.org/abs/2211.14730>.
- Alexis Roger, Gwen Legate, Kashif Rasul, Yuriy Nevmyvaka, and Irina Rish. Small vocabularies, big gains: Pretraining and tokenization in time series models, 2025. URL <https://arxiv.org/abs/2511.11622>.
- Olivier Roy and Martin Vetterli. The effective rank: A measure of effective dimensionality. *15th European Signal Processing Conference (EUSIPCO)*, pages 606–610, 2007.
- Mingtian Tan, Mike A. Merrill, Vinayak Gupta, Tim Althoff, and Thomas Hartvigsen. Are language models actually useful for time series forecasting?, 2024. URL <https://arxiv.org/abs/2406.16964>.
- Kevin Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt. Interpretability in the wild: a circuit for indirect object identification in GPT-2 small, 2023. URL <https://arxiv.org/abs/2211.00593>.
- Malcolm L. Wolff, Shenghao Yang, Kari Torkkola, and Michael W. Mahoney. Using pre-trained llms for multivariate time series forecasting, 2025. URL <https://arxiv.org/abs/2501.06386>.

245 A Additional Tables

Table 6: Representational geometry at Layer 8 for three models processing the same TS input.

Condition	Eff. Rank	PCs for 90%	Speed	Low Freq %
TS → PT	44 ± 29	63	16	34.5%
TS → FT	12 ± 6	15	64	52.3%
TS → RI	35 ± 21	46	78	44.7%
Text → PT	150 ± 8	166	24	21.4%
Text → FT	2.5 ± 1.5	3	355	16.4%

Table 7: Mean principal angle between top-10 PCs of each condition pair at Layer 8. Random baseline: $85.3^\circ \pm 0.4^\circ$.

Pair	Mean Angle	Significance
ts@PT vs ts@FT	60.6°	most aligned
ts@PT vs ts@RI	73.7°	aligned but less
ts@FT vs ts@RI	73.7°	—
text@PT vs text@FT	79.9°	partially shifted
text@PT vs text@RI	85.4°	≈ random

Table 8: Top cross-domain features (PT_FT). Each fires on specific text patterns *and* time series patterns.

Layer	Name	Text activation	TS activation
8	Repetition detector	“3 · 3 · 3 · 3”, “2–3–2 config”	Square waves, periodic oscillations
26	Slope detector	“derivative”, “tangent line”	Rising/falling slopes
4	Metronome detector	“tempo”, “beats”	Sharp periodic spikes
5	Chord detector	“F–Eb–G” progressions	Periodic peaks

246 B Reuse vs. Reinvention: Additional Figures

247 C Causal Circuit Identification

248 The correlational analyses in Section 5 show that finetuning reuses pretrained directions and that cross-domain
 249 features exist. Here we present a preliminary causal analysis: we identify specific model components responsible
 250 for periodic time-series prediction and test what role those components play in language modeling.

251 C.1 Method

252 Since FT_{IO} only trains embedding and LM-head layers, its 28 transformer layers are identical to PT. We zero-
 253 ablate [Wang et al., 2023] each of the 476 components (448 attention heads + 28 MLPs) on FT_{IO} and measure
 254 the loss increase ($\Delta\mathcal{L}$) on periodic time series versus non-periodic controls.

255 **Zero-ablation implementation.** For each attention head, we register a `forward_pre_hook` on the output
 256 projection (`o_proj`) that zeros the head’s 128-dimensional slice of the concatenated 2048-dimensional head
 257 output before projection. For each MLP layer, we register a `forward_hook` that replaces the MLP output with
 258 zeros, so the residual stream passes through unchanged. All ablations are performed under `torch.no_grad()`.

259 **Synthetic time series.** We generate 20 windows (length 512) for each of nine synthetic types. *Periodic*: sine,
 260 square wave, sawtooth, seasonal (trend + oscillation), damped sine. *Control*: white noise, constant, linear trend,
 261 random walk. Each window is independently z -score normalized, clipped to $[-5, 5]$, and uniformly binned into
 262 1024 tokens (matching the FT_{IO} tokenizer). This yields 180 evaluation windows total (100 periodic, 80 control).

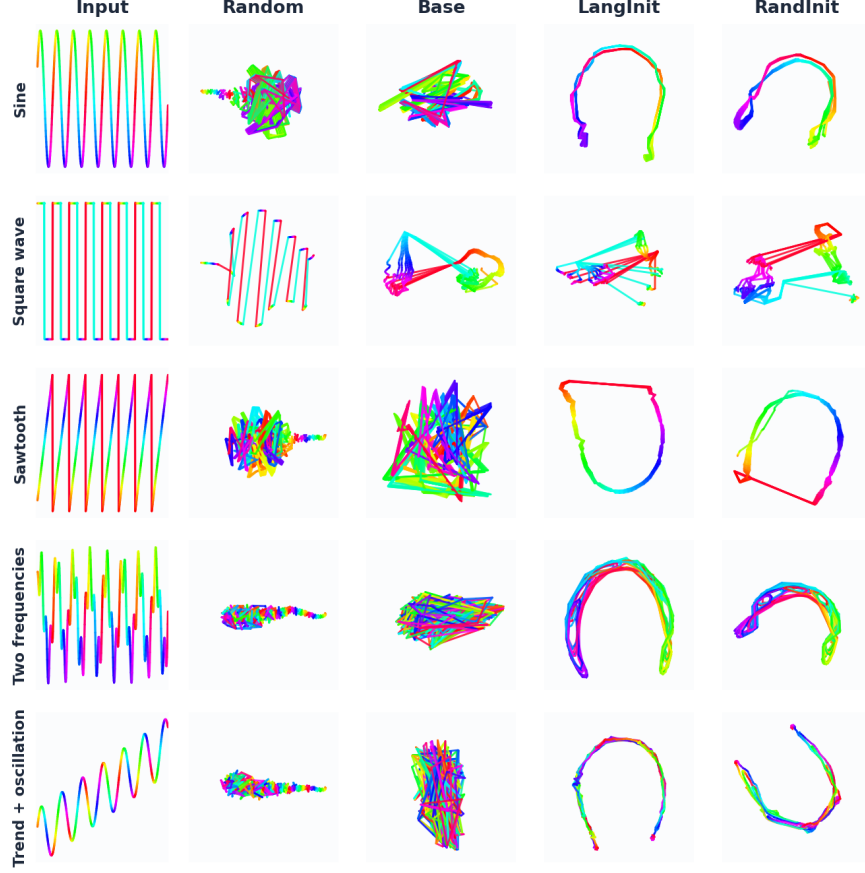


Figure 8: **Hidden-state trajectories for synthetic inputs at Layer 13 (t-SNE).** Same setup as Figure 7 but using t-SNE (perplexity 30) instead of PCA. Despite the different global geometries revealed by PCA, t-SNE shows that LangInit and RandInit develop similar local neighborhood structure: periodic inputs form smooth, phase-ordered curves in both models, confirming that both arrive at functionally equivalent representations through different geometric paths.

263 **Selectivity metric.** For each ablated component we compute:

$$\Delta\mathcal{L}_{\text{periodic}} = \bar{\mathcal{L}}_{\text{ablated}}^{\text{periodic}} - \bar{\mathcal{L}}_{\text{baseline}}^{\text{periodic}}, \quad (3)$$

$$\Delta\mathcal{L}_{\text{control}} = \bar{\mathcal{L}}_{\text{ablated}}^{\text{control}} - \bar{\mathcal{L}}_{\text{baseline}}^{\text{control}}, \quad (4)$$

$$\text{selectivity} = \Delta\mathcal{L}_{\text{periodic}} - \Delta\mathcal{L}_{\text{control}}. \quad (5)$$

264 Components with high $\Delta\mathcal{L}_{\text{periodic}}$ and high selectivity are specifically critical for periodic prediction rather than
265 general-purpose.

266 **Cumulative ablation.** We compose multiple ablation hooks simultaneously using nested context managers.
267 For the pairwise sweep, we test all $\binom{8}{2} = 28$ pairs of the top-8 selective components. For the growing
268 cumulative, we add components one at a time in order of individual $\Delta\mathcal{L}_{\text{periodic}}$. The superadditivity ratio is
269 $\rho = \Delta\mathcal{L}_{\text{combined}} / \sum_i \Delta\mathcal{L}_i^{\text{individual}}$ [Conmy et al., 2023, Elhage et al., 2021]; we use a threshold of $\rho > 1.2$ to
270 account for noise.

271 **WikiText transfer.** We ablate the top-5 circuit components simultaneously on 2,000 WikiText-103 passages
272 (length 512) through the pretrained model (which shares identical transformer layers with FT_{IO}). For each
273 passage we record the per-sequence cross-entropy loss before and after ablation. The 50 most-degraded and 50
274 least-degraded passages are extracted for qualitative analysis.

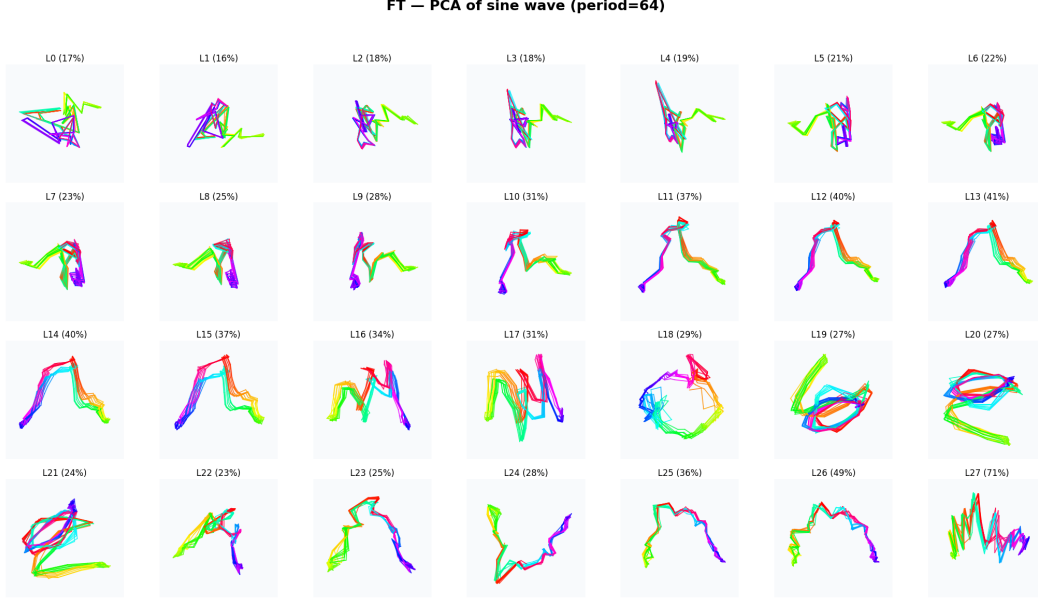


Figure 9: **LangInit: sine wave PCA trajectories across all 28 layers.** Each subplot shows the 2D PCA projection of hidden states from a sine wave (period 64) at one transformer layer, colored by input phase. Percentages show variance explained by 2 PCs. LangInit exhibits layer-specific geometry with varying complexity and moderate PCA variance (17–49%), reflecting the rich, heterogeneous representations inherited from language pretraining.

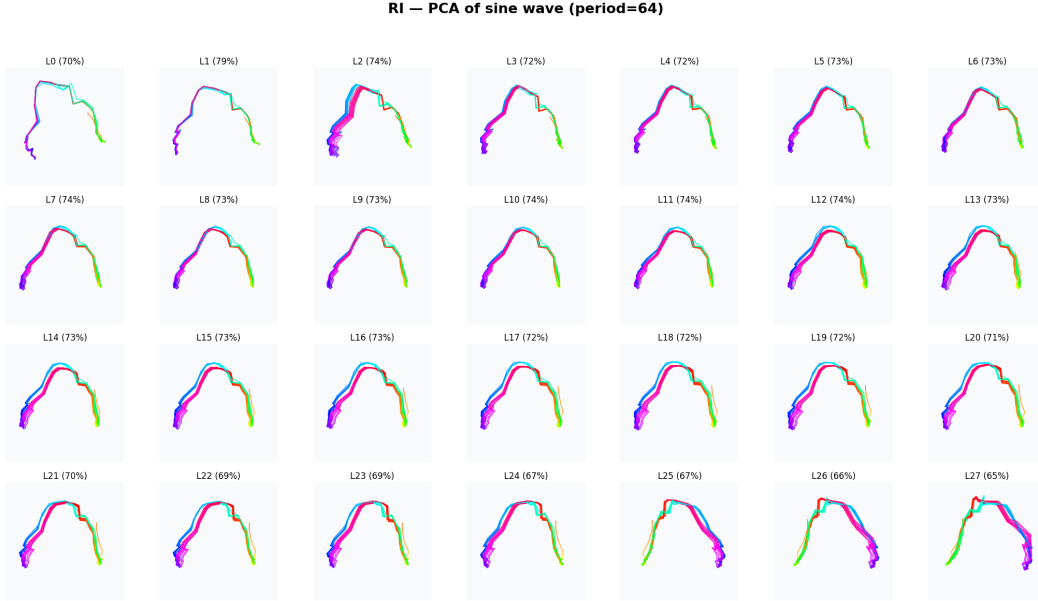


Figure 10: **RandInit: sine wave PCA trajectories across all 28 layers.** Same setup as Figure 9. RandInit produces nearly identical clean arcs at every layer with uniformly high PCA variance (67–79%), confirming that its learned representation is geometrically homogeneous across the network.

275 C.2 Results

276 **Individual component sweep.** Table 9 shows the top-8 components ranked by $\Delta\mathcal{L}_{\text{periodic}}$. The two most
277 impactful are both in Layer 1: MLP_{L1} ($\Delta\mathcal{L}_p = 5.75$, selectivity = 3.44) and head L1H4 ($\Delta\mathcal{L}_p = 4.50$,
278 selectivity = 3.81). Head L20H0 is the third most impactful with the highest selectivity (3.28).

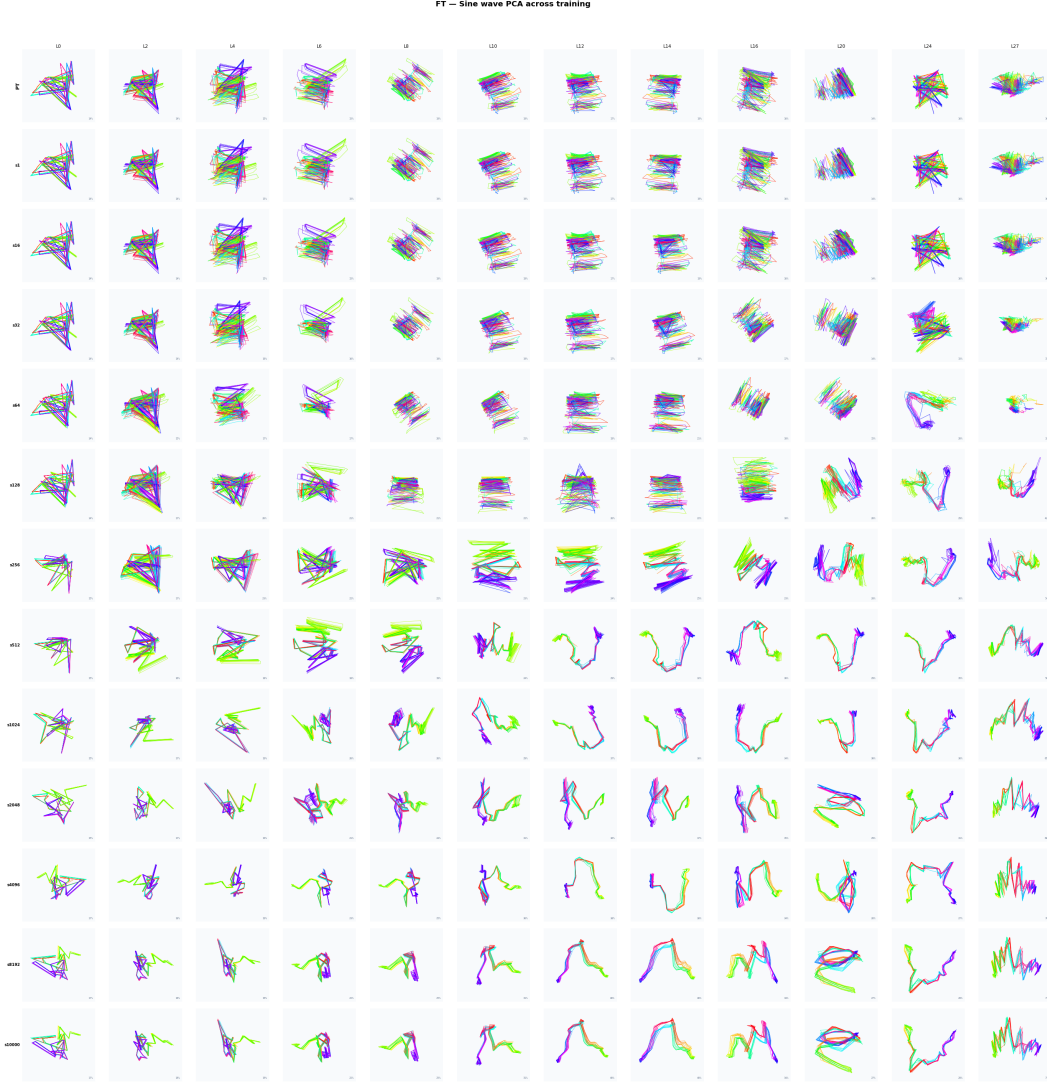


Figure 11: **LangInit: sine wave representations across training checkpoints.** Rows are training steps (top: PT baseline; bottom: final checkpoint at step 10,000), columns are selected layers. LangInit begins from the pretrained model’s complex trajectories and gradually reshapes them into structured loops, with the transition occurring around steps 256–1024.

279 **Cumulative ablation reveals a composed circuit.** The Layer 1 pair (head L1H4 + MLP_{L1}) is strongly
 280 superadditive: ablating them together produces $\Delta\mathcal{L}_p = 15.50$, which is 51% larger than the sum of their
 281 individual effects ($\rho = 1.51$; Table 10). No other pair among the 28 tested exceeds the $\rho > 1.2$ threshold.
 282 Adding head L20H0 maintains superadditivity ($\rho = 1.36$); beyond three components, returns become subadditive
 283 ($\rho < 1$), indicating compensation rather than composition.

284 **The periodicity circuit in language.** We ablate the top-5 circuit components simultaneously on 2,000
 285 WikiText-103 passages through the pretrained model. The circuit ablation causes a mean loss increase of
 286 $\Delta\mathcal{L} = 4.53$ nats—two orders of magnitude larger than any individual head’s WikiText effect (≤ 0.04),
 287 confirming circuit-level interaction.

288 The most-degraded passages ($\Delta\mathcal{L} = 8\text{--}9$; Table 11) are dominated by sequential enumerations and repeating
 289 grammatical templates: biographical sequences (“married to... she gave birth to...”), game mechanics with
 290 parallel clause structure (“sets a task for each stage... this task must be completed... the player with the most”),
 291 Billboard chart progressions, award category listings, and structured Wikipedia sections with repeating headers.
 292 In contrast, the least-degraded passages ($\Delta\mathcal{L} < 2$) are dense, non-repetitive prose: ecclesiastical titles, academic

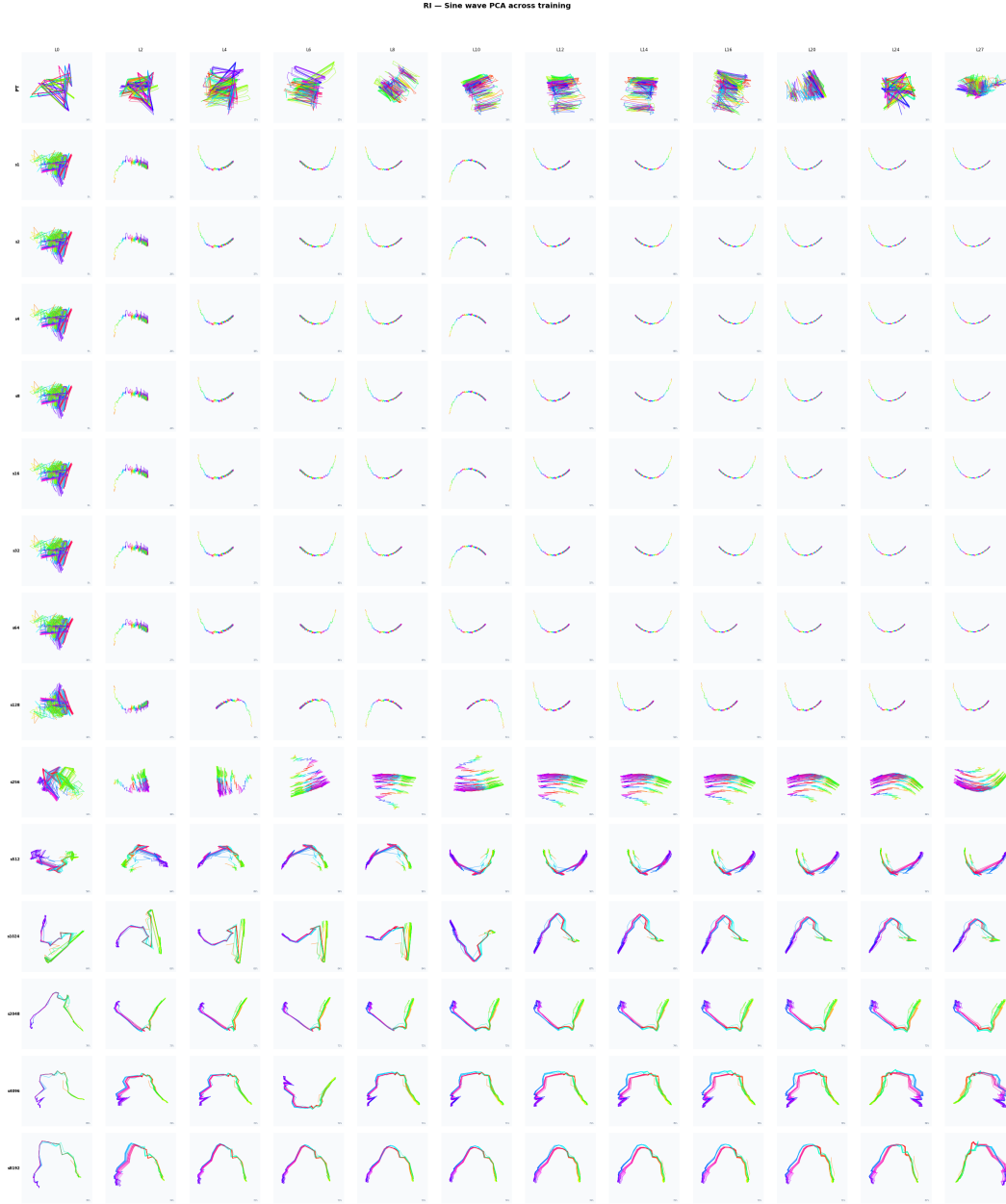


Figure 12: **RandInit: sine wave representations across training checkpoints.** Same setup as Figure 11. RandInit starts from near-empty representations (random weights) and converges to uniform clean arcs by step 2048, with all layers collapsing to the same geometry simultaneously.

293 citations, canonical law, and literary criticism—text where predicting the next token depends on semantic content
 294 rather than structural repetition.

295 This suggests the circuit tracks *sequential repetitive structure*—the same abstract property shared by periodic
 296 time series and repetitive text. However, the evidence is preliminary: the WikiText passages most degraded by
 297 the circuit do not all show an obvious repetitive pattern, and disentangling the circuit’s role in general sequence
 298 modeling from periodicity-specific computation requires further investigation.

Table 9: Top-8 components by $\Delta\mathcal{L}$ on periodic TS under zero-ablation. Selectivity = $\Delta\mathcal{L}_{\text{periodic}} - \Delta\mathcal{L}_{\text{control}}$ isolates periodicity-specific impact. Full sweep: 476 components.

Component	$\Delta\mathcal{L}_{\text{periodic}}$	$\Delta\mathcal{L}_{\text{control}}$	Selectivity
MLP _{L1}	5.75	2.31	3.44
Head L1H4	4.50	0.69	3.81
Head L20H0	3.40	0.13	3.28
Head L13H6	3.35	2.56	0.79
Head L23H6	2.00	0.06	1.94
Head L21H8	1.85	−0.25	2.10
Head L15H3	1.55	0.38	1.18
Head L11H8	1.25	0.56	0.69

Table 10: Cumulative ablation. The Layer 1 pair is strongly superadditive ($\rho = 1.51$); the core circuit saturates at 3 components.

Components	$\Delta\mathcal{L}_{\text{periodic}}$	$\sum \Delta\mathcal{L}_{\text{indiv.}}$	ρ	$\Delta\mathcal{L}_{\text{control}}$
Head L1H4 + MLP _{L1}	15.50	10.25	1.51	2.38
+ Head L20H0 (top-3)	18.55	13.65	1.36	3.69
+ Head L21H8 (top-4)	18.40	15.50	1.19	3.06
+ Head L23H6 (top-5)	18.55	17.50	1.06	3.00
All top-8	18.30	23.65	0.77	3.13

299 D Cross-Domain Feature Analysis via Crosscoders

300 The circuit-level analysis in Appendix C shows that specific components are shared between time-series
301 periodicity prediction and repetitive language modeling. Here we complement that *component-level* analysis
302 with a *feature-level* analysis using crosscoders—sparse autoencoders with a shared encoder applied across
303 domains—to discover individual latent features that fire on both time-series inputs and semantically coherent
304 natural-language passages.

305 D.1 Method

306 **Setup.** We train one *linear crosscoder* per transformer layer of Qwen3-0.6B. Each crosscoder has a
307 shared encoder $\mathbf{W}_{\text{enc}} \in \mathbb{R}^{1024 \times 4096}$ followed by Top- K sparsity ($K=64$), and two per-domain decoders
308 $\mathbf{W}_{\text{dec}}^{(\text{PT})}, \mathbf{W}_{\text{dec}}^{(\text{FT})} \in \mathbb{R}^{4096 \times 1024}$ (12.6 M parameters total, float32). The encoder is applied independently to each
309 domain’s hidden states; because the weights are shared, the same latent feature can fire on both pretrained (PT)
310 and fine-tuned (FT) inputs.

311 **Domains.** *PT*: Time series formatted as space-separated decimal strings (e.g., 0.123 -0.456 1.789 . . .),
312 tokenized by the Qwen3-0.6B tokenizer. Sub-tokens corresponding to each timestep value are mean-pooled to
313 produce one 1024-dim vector per timestep. *FT*: The same time series normalized per-window (z -score), clipped
314 to $[-5, 5]$, and uniformly binned into 1024 tokens for the fine-tuned model.

315 **Training.** Input: 3,000 windows ($T=512$) from GiftEval [?], precomputed as memory-mapped hidden-state
316 arrays. Loss: per-domain MSE between normalized inputs and decoder reconstructions, summed over PT and
317 FT. Dead-feature recovery (AuxK; top-64 among features firing $<1\%$ over the last 1,000 steps, weight $\frac{1}{32}$, active
318 after step 1,000). AdamW, $\text{lr}=3 \times 10^{-4}$, 500 warmup steps, cosine decay over 10,000 steps. Early stopping after
319 1,500 steps without validation improvement (minimum step 2,000).

320 **Feature analysis pipeline.** After training, 50,000 time-series windows are encoded; for each of the 4,096
321 features we record the PT and FT firing rates (fraction of windows with non-zero activation). Features firing $\geq 1\%$
322 in both domains are labeled PT_FT and ranked by balance = $\min(\text{rate}_{\text{PT}}, \text{rate}_{\text{FT}})$. For the top-30 balanced
323 features, we extract the 10 highest-activating time-series windows and 10 highest-activating WikiText-103
324 passages (30,000 sequences, 512 tokens each, encoded through the PT model and the shared crosscoder encoder
325 with separate normalization statistics).

326 D.2 Results

327 Layers 7–10 are the sweet spot for shared cross-domain features (Table 12). Early layers (0, 6) and late layers
 328 (13, 20) show almost no balanced features. Below we present four features with the clearest cross-domain links:
 329 each fires on a specific time-series pattern *and* on thematically coherent WikiText passages.

330 **Feature 1712 (Layer 10) — Quantitative magnitude transitions.** PT firing rate: 50.4%, FT: 41.8%,
 331 WikiText: 14.1%. This feature detects step changes and regime shifts in time series—values that jump from a
 332 near-zero baseline to a sustained high plateau—and fires on text passages dense with numerical measurements
 333 and unit conversions (Figure 13).

Feature 1712 (Layer 10) — Quantitative magnitude transitions

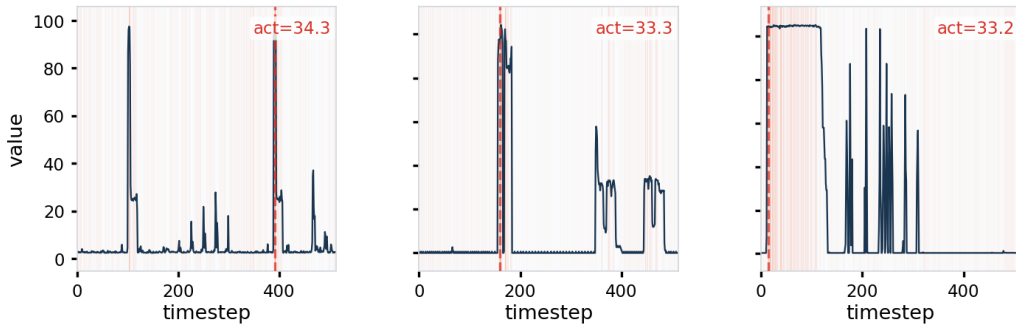


Figure 13: **Feature 1712 (Layer 10): Quantitative magnitude transitions.** Top-3 activating time-series windows. Blue: raw signal; red dashed: peak-activation timestep; orange shading: activation intensity. All three windows share a sudden jump from a low baseline to a high-magnitude plateau.

334 Top-5 WikiText passages at the peak-activation token:

- 335 1. (*act=14.1*) "... becoming later in the year by about two days every 243-year cycle. Transits usually occur in
- 336 pairs, on nearly the same date eight years apart."
- 337 2. (*act=13.8*) "In practice, forward premiums and discounts are quoted as annualized percentage deviations
- 338 from the spot exchange rate..."
- 339 3. (*act=13.7*) "Wages are reflective of the type of jobs available locally, including higher than average employ-
- 340 ment in manufacturing and the public sector. The working age population of the town in 2011..."
- 341 4. (*act=13.7*) "So for americium-241, the resistivity at 4.2 K increases with time from about 2 $\mu\text{Ohm}\cdot\text{cm}$ to
- 342 10 $\mu\text{Ohm}\cdot\text{cm}$ after 40 hours, and saturates at about 16 $\mu\text{Ohm}\cdot\text{cm}$..."
- 343 5. (*act=13.7*) "Falcon's Fury can theoretically accommodate 800 riders per hour. Carbon-fiber wings buttress
- 344 each end of a group of seats..."

345 The shared representation encodes *quantitative magnitude and transition*: literal level shifts in time series, and
 346 passages dense with measurements, unit conversions, and numerical comparisons in text.

347 **Feature 2469 (Layer 9) — Tropical weather systems.** PT: 29.9%, FT: 20.7%, WikiText: 12.6%. Fires
 348 on volatile, regime-switching time series and exclusively on tropical cyclone and hurricane narratives in text
 349 (Figure 14).

350 Top-5 WikiText passages:

- 351 1. (*act=8.7*) "... the mean locus of formation shifts westward to the Caribbean and Gulf of Mexico, reversing
- 352 the eastward progression of June through August. Wind shear from westerlies increases substantially through
- 353 November..."
- 354 2. (*act=8.3*) "... due to a combination of very high wind shear and dry air. By October 17, most of the deep
- 355 convection associated with the system dissipated; however, a brief decrease in wind shear allowed Omar to
- 356 re-strengthen..."
- 357 3. (*act=8.1*) "The wave continued westward and related thunderstorm activity increased during the following
- 358 week. The convective system organized into Tropical Depression Twenty-E on September 28..."
- 359 4. (*act=8.1*) "A tropical wave moved across the northeast Pacific Ocean and formed a tropical depression south
- 360 of Mexico on October 16. It strengthened at a moderate pace and reached hurricane intensity on October 18."

Feature 2469 (Layer 9) — Tropical weather systems

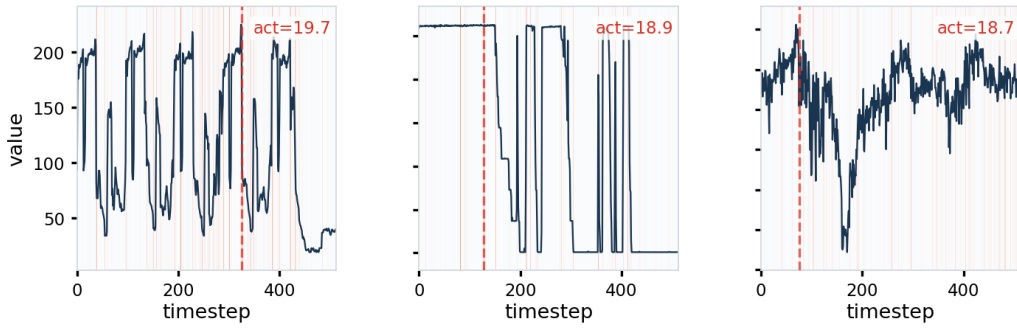


Figure 14: **Feature 2469 (Layer 9): Tropical weather systems.** Top-3 activating windows. The left window shows high-variance oscillations with abrupt drops; the middle and right show diverse volatile patterns with regime switching.

361 5. (*act*=7.9) "... formation of Typhoon Chanchu in the western Pacific enhanced convective activity over the
 362 Bay of Bengal. By April 22, a trough developed along an axis from the southern Bay of Bengal eastward to
 363 the Andaman Sea."

364 The time-series patterns—volatile signals with sudden regime changes—mirror the physical phenomena de-
 365 scribed in the text: tropical storms that intensify, weaken under wind shear, and shift track.

366 **Feature 3888 (Layer 7) — Naval battle events.** PT: 18.5%, FT: 26.3%, WikiText: 10.5%. Fires on sharp
 367 isolated spikes in otherwise stable time series and on naval/military battle narratives with precise timestamps
 368 (Figure 15).

Feature 3888 (Layer 7) — Naval battle events

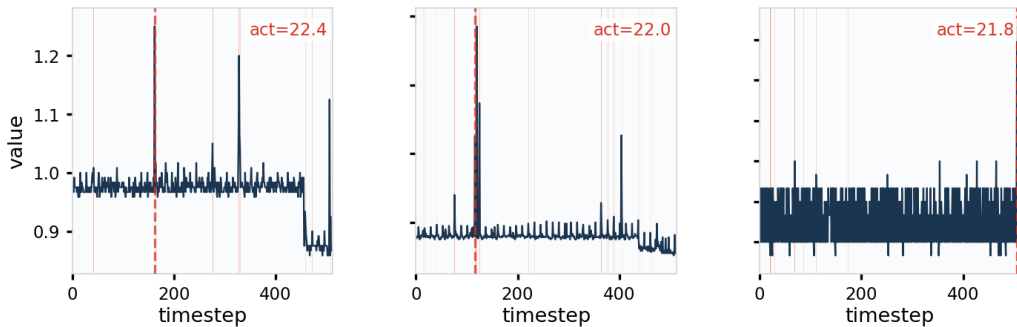


Figure 15: **Feature 3888 (Layer 7): Naval battle events.** Top-3 activating windows. Each shows a low-variance baseline punctuated by sharp spikes at the peak-activation timestep (red dashed line).

369 Top-5 WikiText passages:

- 370 1. (*act*=10.1) "King George V had only 32 percent of her fuel left while Rodney had only enough fuel to
 371 continue the chase at high speed until 8:00 the following day. Admiral Tovey signalled his battlegroup..."
- 372 2. (*act*=9.3) "At 7:20 on 19 July, the destroyer force spotted and was spotted by a pair of Italian light cruisers;
 373 Giovanni dalle Bande Nere and Bartolomeo Colleoni, which opened fire seven minutes later."
- 374 3. (*act*=9.2) "Shortly before 16:00 the battlecruisers of I Scouting Group encountered the British 1st Bat-
 375 tlecruiser Squadron under the command of Vice Admiral David Beatty. The opposing ships began an
 376 artillery..."
- 377 4. (*act*=9.2) "The eastern wind was not communicated to the aircraft, but was 270°, varying from 20 to 40 knots
 378 (37 to 74 km/h). The take-off started at 14:42:43..."
- 379 5. (*act*=9.1) "... torpedo boat attacks and at 07:30, Burrough sent Eskimo and Somali back to help Manchester
 380 but they arrived too late, took on survivors..."

Both modalities encode *sudden, precisely-located events*: an anomalous spike at a single timestep in time series, and a precisely-timestamped combat event in text.

Feature 2567 (Layer 8) — Missing / null data. PT: 35.9%, FT: 36.8%, WikiText: 9.4%. This feature fires exclusively on NaN/missing time-series data (all 10 top-activating windows are entirely NaN, with peak activation 40.3) and on <unk> tokens and incomplete references in text.

Top-5 WikiText passages:

1. (*act*=19.8) "... at the Royal Navy School of Flight Deck Operations at RNAS Culdrose. The following is an incomplete list of some of the surviving aircraft."
2. (*act*=18.5) "<unk>, <unk>, <unk>, <unk>, <unk>, Ulaid. Slightly later major groups included the Con-nachta, <unk>, <unk>. Smaller groups included the <unk>..."
3. (*act*=18.0) "... he encountered bad weather, forcing him to return to Japan with heavy damage. Without waiting for Vizcaino, another ship—built in Izu by the Tokugawa shogunate..."
4. (*act*=17.8) "Luke 9: <unk>-<unk> — καὶ <unk>, <unk> <unk> <unk> πνεύματος <unk> <unk>..."
5. (*act*=17.7) "... whom he married in the late 250s when she was 17 or 18 years old. The number of children Odaenathus had with his first wife is unknown and only one is attested."

The model represents *absent information* identically across modalities: NaN values in time series and <unk> tokens in text both occupy the same region of representation space.

D.3 Discussion

The crosscoder analysis reveals that layers 7–10 contain specific, thematically coherent features shared between time-series and language representations. The most interpretable features tend to have moderate balance (15–45%) rather than the highest balance (>50%), which corresponds to generic backbone features. The four features documented above encode distinct representational primitives—*quantitative magnitude transitions*, *volatile regime switching*, *punctual disruption*, and *absent information*—that manifest as both time-series patterns and specific text genres.

This complements the circuit-level finding from Appendix C: the periodicity circuit’s components operate at specific layers (L1, L20), while the crosscoder features are concentrated in the intermediate layers (7–10) where the PT and FT representations are most aligned. Together, these analyses suggest that LangInit exploits both circuit-level and feature-level structure from language pretraining when processing time series.

409 CHECKLIST TODO

Table 11: WikiText-103 passages most and least degraded by ablation of the periodicity circuit (top-5 components). Mean $\Delta\mathcal{L} = 4.53$ across 2,000 passages.

$\Delta\mathcal{L}$	Passage excerpt
<i>Top 20 most degraded</i>	
9.19	"...Townsend has been married to Tracy Turner, his girlfriend since he was 19. She gave birth to thei[r]..."
8.88	"...Preston winger Will Hayhurst, a Republic of Ireland under-21 international, was signed on a one-month loan..."
8.88	"...with the NHK Symphony Orchestra, but cancelled both deals upon Mwanga's return from Japan. Mwanga immediately quit..."
8.88	"...his/her final score on the song, with money being awarded in Guitar Hero World Tour. The games have also added..."
8.75	"...Viscount, sets a task for each stage. This task must be completed before the player can continue to another map..."
8.50	"...The player character is Michel Ardan, an eccentric and intrepid French scientist who is enthusiastic, daring..."
8.38	"...Best Music Video, Long Form. In 1998, the categories were retitled Best Short Form Music Video, and Best Long..."
8.31	"...on the Billboard Hot 100 twenty-nine, the highest U.S. entry among all singles released from the album..."
8.31	"...long run, with The A.V. Club attributing much of the show's early success to the character..."
8.25	"...Yankovic recorded 'Here's Johnny', a parody of 'Who's Johnny' by El DeBarge. The song, a loving ode to..."
8.19	"...The music video of 'Crazy in Love', released in May 2003, was directed by Jake Nava and filmed..."
8.19	"...Finishing with the worst record in the NHL, Columbus had the best chance of receiving the first overall pick..."
8.13	"...while one in the Pyramid Texts says the name is based on words shouted by Osiris..."
8.06	"...the album Daydream most resembles in its emphasis on R&B grooves. Tucker specifically complimented 'One Sweet Day'..."
8.00	"...similar to Konami's Guitar Freaks and to a lesser extent Harmonix's previous music games such as Frequency..."
8.00	"...he 'hit the wall with play-along music games', and challenged the game makers to explore other ways..."
7.94	"...saying that he 'was upset. But when you see the talent that was there, it was an honour just to be in the final'..."
7.94	"...Flags indicate national team as defined under FIFA eligibility rules. Players may hold more than one non-FIFA..."
7.94	"...at the Television Critics Association summer media tour in Beverly Hills, California..."
7.94	"...co-wrote and produced a song with Kenneth 'Babyface' Edmonds, with whom she had collaborated on Music Box..."
<i>Top 10 least degraded</i>	
0.83	"...Porto e Santa Rufina; Sub-dean of the Sacred College of Cardinals; prefect of the S.C. of the Good Government..."
1.48	"...Cardinal-Priest of SS. Giovanni e Paolo; Grand penitentiary; prefect of the Congregation for the correction..."
1.59	"...From the Jewish Question to the Jewish State: An Essay on the Theory of Zionism (thesis), Princeton Univ..."
1.77	"...called Saprang's transfer a 'demotion' and a 'punishment.' However, Saprang himself claimed that he did not..."
1.82	"...the Society of Jesus, provided he observed the canon law; and that it was desirable that the pope should..."
1.89	"...the abnormal excess of white blood cells in people with the clinical syndrome described by Velpeau and Bennett..."
1.89	"...The protagonist sounds like a 'colonial administrator', and his reference to seeking a newer world echoes..."
1.94	"...Mbaruk's nephew, Mbaruk bin Rashid, refused to acknowledge the appointment of a new leader..."
1.98	"...00 works were published underground over the course of the war. Literary discussions were held..."
2.02	"...Although the poem was defended by a few critics, E.C. Pettet returned to the argument that the poem lacked..."

Table 12: Cross-layer feature balance. “Best balance” is the highest $\min(\text{rate}_{\text{PT}}, \text{rate}_{\text{FT}})$ among all 4,096 features. Layers 7–10 concentrate almost all balanced features.

Layer	Best balance	# Features >30% bal.	Train loss
0	2.1%	0	0.016
6	29.4%	0	0.050
7	57.0%	10	0.052
8	52.1%	6	0.062
9	51.2%	7	0.067
10	55.7%	5	0.057
11	47.1%	4	0.062
12	47.9%	3	0.062
13	50.0%	1	0.059
20	2.0%	0	0.120